

Do High-Rank Dota 2 Players Move Differently?

Predicting Skill from Spatio-Temporal Movement

SIADS 699 Capstone Report · Team DotA Science
University of Michigan School of Information

1. Project statement

If you watch a Dota 2 replay from a Legend player (high skill rank) and another from someone still climbing out of Guardian (low skill rank), you might imagine the map might look totally different at a glance. But, we'll spoil the secret from the start, it's not that different. Heroes still lane, rotate, and fight over the same objectives. Because of that, the question this project asks is pretty narrow and more than a fair bit stubborn: can the way a player moves, stripped of gold, items, and kill counts, still carry information about skill?

Well, for starters let's set the stage. Dota 2 is a five-versus-five game where each match lasts roughly thirty to fifty minutes. Public ranked matches expose a medal system that roughly tracks ability. Prior work has shown that team positioning and zone changes differ across skill tiers (Drachen et al., 2014), and that in-game state can predict match outcomes (Hodge et al., 2017). Much less work tests whether individual movement trajectories alone are enough to separate higher- and lower-ranked players in ordinary public games.

Our goal was practical: build a pipeline from public replays to movement features, then test whether those features predict skill labels better than chance. Success, for us, was not a leaderboard score. It was a clear answer about how large the movement signal is, what muddies it, and whether the result is honest enough to put in a coaching or analytics product.

2. Related work

Well, first off, Drachen et al. (2014) measured zone changes and within-team distances across four skill tiers in Dota 2 and found that higher-skill teams stayed more compact and changed zones more often. That paper is the closest to our project: it treats space and time as primary signals of skill, not as decoration around economy stats.

Then we have Hodge et al. (2017) that studied live win prediction in mixed-rank and professional matches. Their focus was outcome, not individual skill, but the work matters here because it showed that public and professional data can be combined with care, and that feature choice dominates algorithm choice for this domain.

Lastly, more recent pipelines for skill identification (for example Pengmatchaya and Natwichai, 2024) fold spatiotemporal features into broader feature sets that also include end-game statistics and harassment tactics. Those models perform well, which is encouraging, but they also leave open a cleaner question: how far can movement go on its own before the rest of the scoreboard is invited back in?

3. Data and methodology

3.1 Data collection

We collected ranked All-Draft matches through the OpenDota API, downloaded the associated replay files, and parsed player positions with the Clarity replay library. After filtering for valid demos and usable rank information, the analysis set contains 533 matches and 4,993 labeled players. Most matches come from European servers (training pool), with a held-out regional slice intended to test whether any signal travels. Replays are large; a single decompressed demo can exceed 150 MB, so collection and parsing were treated as a first-class engineering problem rather than a side detail.

Each match was assigned a high or low skill label from the lobby's average rank tier. High lobbies sit roughly at Legend and above; low lobbies sit below that band. Individual rank_tier values still overlap across those labels, which is expected under matchmaking and becomes important later when we interpret weak average differences.

3.2 From coordinates to features

Raw map coordinates are team-asymmetric: Radiant and Dire spawn on opposite corners. A model trained on raw x and y can learn side instead of skill. We therefore transform every position into a team-relative frame where own_depth runs from a player's own fountain (0) toward the enemy fountain (1), and own_lateral measures distance from the mid-lane diagonal. Lanes are cut from own_lateral into top, mid, and bottom bands.

We aggregated trajectories in two ways. First, fixed clock phases used for modeling features (early laning, mid game, late game). Second, equal thirds of each match's own duration, which supports the exploratory comparison plots below. For each player-match we compute quantities such as mean depth, speed, idle share, path efficiency, and time spent in each lane-phase combination. A simple death heuristic masks periods after a teleport-like return to fountain so respawn jumps do not dominate distance statistics.

It also bears mentioning that the feedback from early on correctly flagged that position extraction is the linchpin of the project and that compute cannot be treated casually. We prototype-parsed with Clarity before scaling, stored intermediates in Parquet, and kept the feature build match-by-match to control memory. Those choices are unglamorous. They are also why the rest of the analysis is possible.

3.3 Modeling approach

The primary model is an XGBoost classifier predicting high versus low skill from movement features. We compare several feature families rather than searching every subset: all movement features, movement without speed, movement without lane-share features, and a smaller core trajectory set. Where possible we also inspect models that include hero identity, because hero role can mimic skill in raw movement (supports and carries do not patrol the map the same way).

Evaluation uses stratified train-test splits for rapid iteration, with the understanding that grouped validation by match is the more principled check against lobby leakage. Earlier exploration showed that a large share of rank variance sits at the match level, so features built from the other nine players are treated as risky and mostly avoided in the self-focused feature sets.

4. Evaluation strategy

We defined success in three layers.

First, descriptive separation: do high and low groups differ on movement summaries in the expected direction? Faster movement and less idle time for higher ranks would count as weak but coherent evidence.

Second, predictive signal: can a movement-only model beat a naive baseline on held-out players? AUC(Area Under the Curve) meaningfully above 0.5 is the minimum bar. Large gains from adding hero identity without movement would suggest the movement story is mostly role confounding.

Third, interpretability and honesty: SHAP(SHapley Additive exPlanations) summaries should not collapse to a single accidental proxy, and the report should state when effects are small. A successful outcome, in that sense, includes a negative or modest result if the evidence supports it.

5. Results

The cleanest finding is also the least dramatic. Across equal thirds of each match, high and low lobbies use the three lanes in almost the same proportions. Early game is split between the side lanes; later game compresses toward mid. Rank does not redraw that schedule.

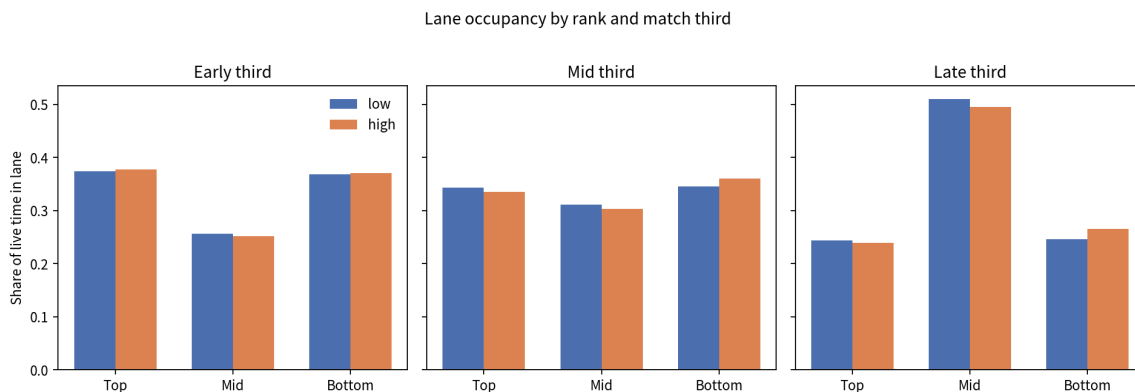


Figure 1. Share of live time spent in top, mid, and bottom lanes for high and low skill lobbies, split into early, middle, and late thirds of each match. High and low lobbies spend similar fractions of time in each lane across the match.

Continuous movement metrics move in a more interesting direction. Higher-rank lobbies show slightly higher mean speed and slightly lower idle share, especially in the middle third of the match. Mean depth, our crude aggression proxy, barely budes. The gaps are real but tiny: on the order of a few speed units and one to three percentage points of idle time.

Higher-rank lobbies move a bit more and idle a bit less

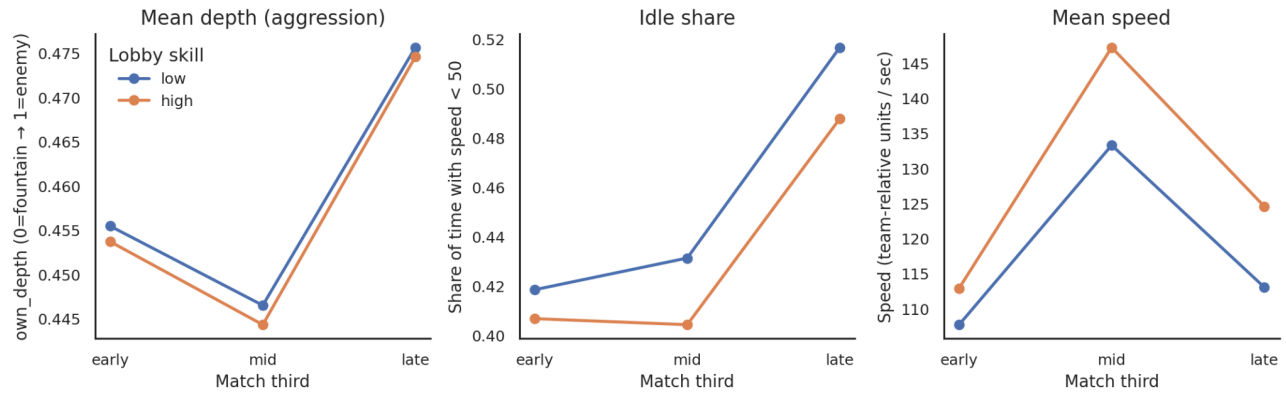


Figure 2. Mean depth, idle share, and mean speed across match thirds for high and low skill lobbies. Speed and idle time separate ranks more than depth does. Speed is a bit higher and idle share a bit lower for high-rank lobbies; mean depth barely moves.

Variable	Definition
Mean Depth	Mean of own_depth over a player's live samples (0 = own fountain, 1 = enemy fountain)
Mean Speed	Mean of per-sample displacement ÷ elapsed time between samples
Idle Share	Fraction of live samples whose speed falls below a fixed threshold

Positions are sampled every two seconds. The 'live' condition filters out samples identified by the death heuristic (instant returns to the fountain and the subsequent eight seconds). All three metrics are computed per player-match and averaged within each skill group. Each match is divided into three equal time intervals (match thirds) based on its total duration.

Map density plots tell a similar story. Early-game occupancy for high and low lobbies concentrates in the expected laning corridors. There is no obvious alternate highway that only high-rank players use. If skill lives in movement, it is more likely in timing, efficiency, and response to events than in a permanently different preferred region of the map.

Where players spend time early in the match

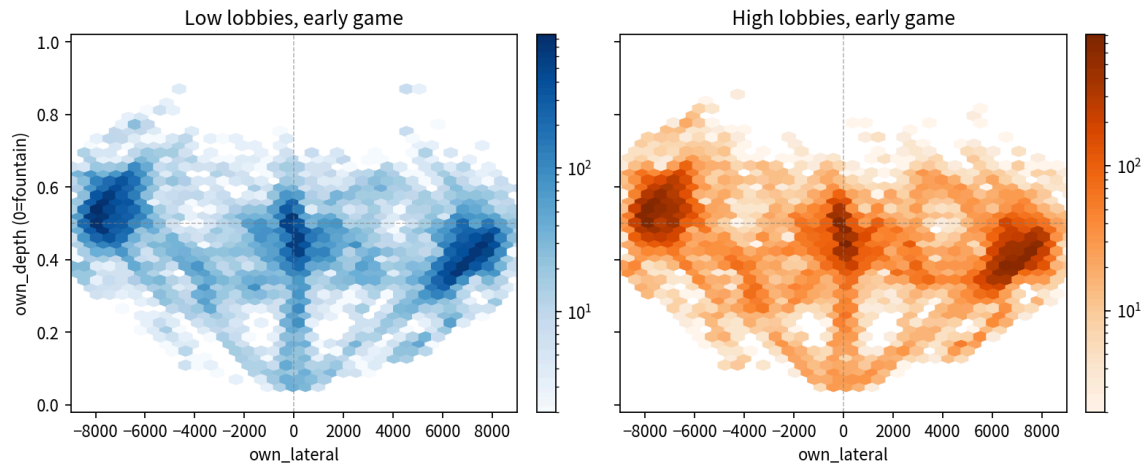


Figure 3. Early-game position density in team-relative coordinates for low and high skill lobbies. Darker cells mean more time spent in that region of the map. Early-game density concentrates in the same laning corridors for both groups.

Part of the modest separation is structural. Match-level labels group players who are not identical in rank. Figure 4 shows individual rank_tier values inside high and low labeled matches. The centers differ, but the tails overlap. Any analysis that treats every player in a high lobby as interchangeable will understate individual differences and overstate lobby composition.

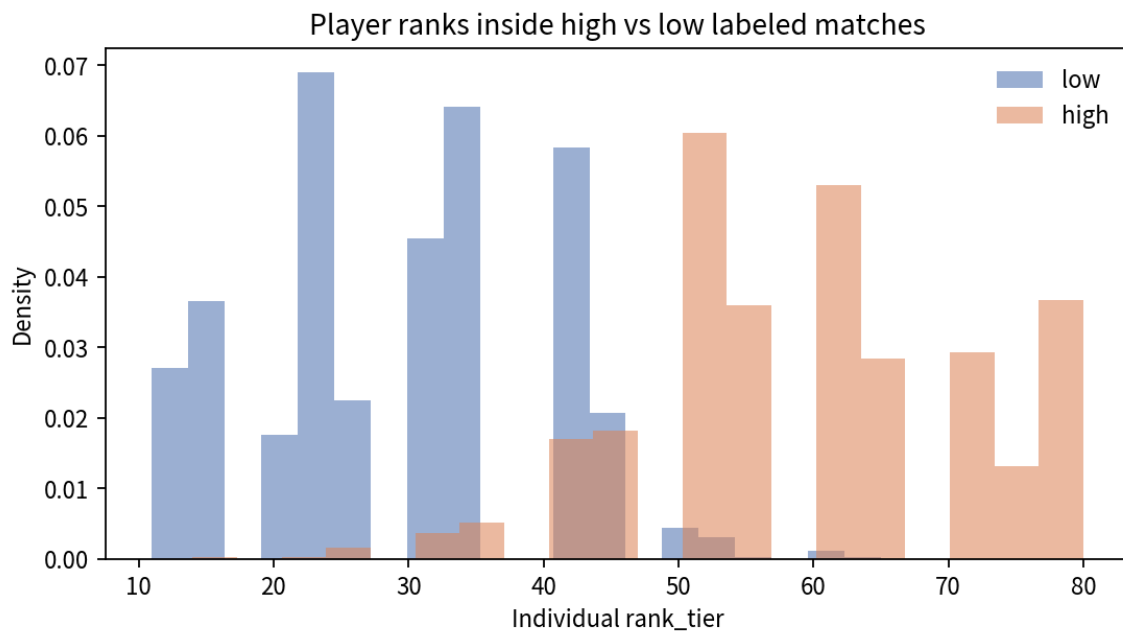


Figure 4. Distribution of individual player rank_tier inside matches labeled high or low from lobby average rank. Overlap is visible even though the means are well separated. Match-level high/low labels still contain overlapping individual ranks, which softens average gaps.

Modeling results track the descriptive picture. On a held-out 20% player split, an XGBoost classifier trained only on movement features reached an AUC of 0.79 and accuracy of 0.73. The best feature sets were trajectory summaries without lane-share features, or a smaller core trajectory set (both AUC $\approx$ 0.79). A

minimal depth/path/idle set performed much worse (AUC 0.62), which supports the idea that richer movement summaries matter more than a few aggregate scalars. Lane-share features did not help once trajectory features were present. These numbers beat chance, but they are not strong enough on their own for a ranking product. The split was not grouped by match, so some lobby-level leakage may remain. Adding hero information can improve fit but that improvement is a warning light as much as a win, because it shows how easily role substitutes for skill in spatial data.

6. What the numbers mean

Overall, you can say that high-rank players in our sample do not occupy the map differently. They occupy the same map a little more actively. They idle less. They move a bit faster. Those are the movement signatures that survive averaging.

That pattern is compatible with Drachen et al.'s (2014) broader claim that spatial behavior tracks skill, while also showing why public-match individual prediction is harder than tier-averaged team analysis. Public lobbies are mixed. Roles differ. Two-second samples blur micro-mechanics. Under those conditions, for us, weak average effects are not a failed parse. They are a plausible estimate of how much signal remains after.

For a coaching tool, which was our frame of reference, the implication is conservative. Movement features may help flag extreme inactivity or odd pathing once hero and role are controlled. But, they are unlikely, with just movement, able to certify that one Legend player is better than another. Still, for research, the more useful contribution is the pipeline itself: team-relative coordinates, phase-aware aggregation, and an explicit treatment of lobby leakage.

7. Broader impacts and ethics

Who is affected if this kind of model works? Players, first. Automated skill inference from replays could feed coaching products or matchmaking experiments. Analysts and teams could use trajectory summaries to review habits that manual review might miss. Now, to be fair, platform holders already collect richer telemetry than we used (by a large margin); academic pipelines mostly rediscover slices of that private view.

Overall, for this kind of work, one ethical concern is misclassification with social consequences. A model that tags a player as low-skill from movement alone can be wrong for boring reasons: an unusual hero pool, a support role that looks passive on a heatmap, or a match spent intentionally sacrificing space. Publishing or productizing those scores without uncertainty and context risks harassment, unfair bans, or coaching advice that simply tells supports to play like carries. A second concern is privacy. Replays are public, but linking movement profiles to accounts still creates a behavioral fingerprint. We used public ranked data and focused on aggregate patterns, which is necessary but not sufficient if the same methods are later applied at scale.

8. Limitations and response to mentor feedback

Our mentor's review pushed on the right weak points: replay parsing had to be proven early, compute could not be hand-waved, success was out of scope, feature aggregation needed a concrete design, and evaluation should isolate what movement adds. We responded by locking the parser path on Clarity, storing compressed intermediates, focusing on skill rather than success, defining phase and zone rules in a written feature spec, and running ablations instead of an exhaustive combinatorial search over feature drops.

Still, our remaining limitations are honestly quite substantial. Death and respawn handling is heuristic. Vision, abilities, and items are absent. Equal-time thirds are only a proxy for true game phases. Match-level labels blur individual skill. None of those caveats disappear because the plots look tidy.

9. Statement of work

As Team DotA Science we just had a tight vision and narrative we wanted to explore. The breakdown of roles is as follows:

Gabriel Mejia: project framing, feature-spec design, exploratory analysis of phase and lane structure, report writing, and integration of mentor feedback into the evaluation plan.

Changwoo Eric Kim: Built a data pipeline using the OpenDota API to collect ranked replays at scale, archive raw replay files to Google Cloud Storage under a versioned storage layout, parse player positions, and transform them into an analysis-ready dataset, including validation.

Sung Jin Bae: initialized baseline modeling and SHAP(SHapley Additive exPlanations) value calculation, and supported overall visualization improvements. Also, arranged regular meetings, scheduled check-in with instructors and shared logs after meeting through slack for project managing.

Modeling and visualization support: XGBoost ablations, SHAP(SHapley Additive exPlanations) value inspection, and figure generation for the descriptive comparisons shown above. Final model choices and the conservative interpretation of effect sizes were reviewed by all three of us (Gabriel, Sung Jin, Changwoo).

10. Conclusion

Movement in Dota 2 is not silent about skill, but it's certainly not loud either. In public ranked replays, higher-rank lobbies move a little more and rest a little less, while lane preferences and average depth remain stubbornly similar. The honest product of this project is a reproducible path from replay files to team-relative movement features, plus evidence that any claim about skill-from-motion needs hero controls, leakage checks, and a tolerance for small effects. Which is to say that this is a narrower victory rather than a flashy accuracy number but it is also simply what the data actually supports.

References

- Drachen, A., Yancey, M., Maguire, J., Chu, D., Wang, I. Y., Mahlmann, T., Schubert, M., & Klabjan, D. (2014). Skill-based differences in spatio-temporal team behaviour in Defence of the Ancients 2 (DotA 2). *IEEE Conference on Computational Intelligence and Games*.
- Hodge, V. J., Devlin, S., Sephton, N., Block, F., Drachen, A., & Cowling, P. I. (2017). Win prediction in esports: Mixed-rank match prediction in multi-player online battle arena games. *arXiv:1711.06498*.
- Pengmatchaya, M., & Natwichai, J. (2024). Identifying player skill of Dota 2 using machine learning pipeline. *Discover Artificial Intelligence*, 4, Article 39. <https://doi.org/10.1007/s44163-024-00139-y>

Appendix: What we left out

There's a lot to say here given our original scope. Firstly, several threads never earned a full section in the main report. We experimented with stricter individual rank bands (for example rank 60 and above versus 30 and below) to reduce label noise; the direction of the speed and idle effects stayed the same. We considered multiple ways of segmenting the match temporally but that does not change the aggregated features. An exhaustive combinatorial search over feature subsets was attempted early and abandoned after it became clear the search space was millions of models wide. Sensible ablations replaced it. Economy-only baselines remain a natural extension if gold and XP(Experience Point) timelines are parsed from the same replays. Analysing movement for each hero based on skill. Policy-style or imitation-learning frames for movement were discussed as future work and intentionally kept out of scope for the class timeline.

The appendix does not do justice to all the interesting possibilities we discussed but these are ripe venues for future work for anyone interested in furthering the work.